

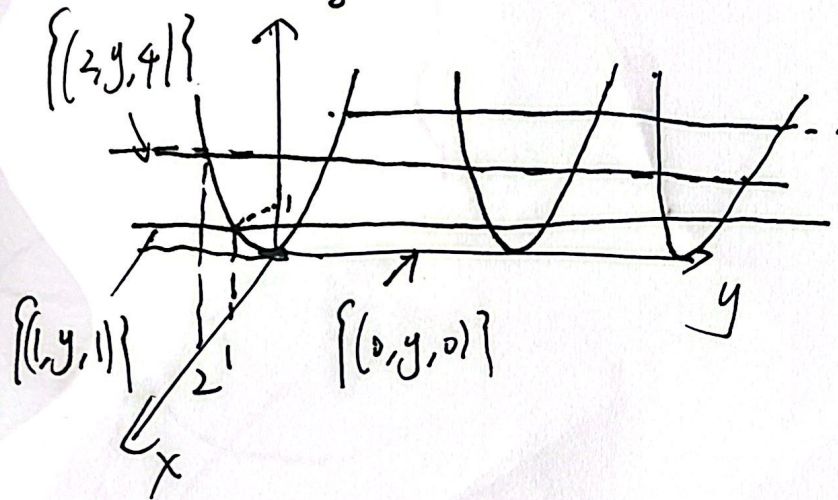
Sep 16.

Cylinders

Consider a function $f(x,y) = x^2$.

Q: What happens to $f(x,y)$ when we fix x and move y ?

$$f(x,y_1) = x^2 = f(x,y_2) \quad \text{for any } y_1, y_2.$$



The key point is that y doesn't appear in the definition of f . (algebraic way to identify if an object is a cylinder).

The key geometric property of a cylinder is that it is made of a family of lines pointing in the same direction.

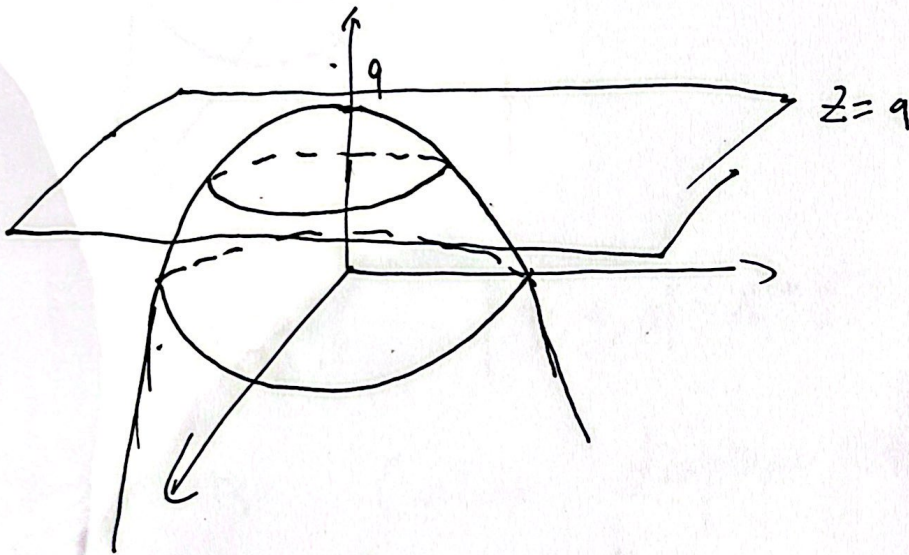
In the above example, the graph of $f(x,y) = x^2$ is the union of lines $\ell_{x_0} = \{(x,y,z) : z = x_0^2\}$ for $x_0 \in \mathbb{R}$

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Equation	Surface	Direction of Extension
$z = x^2$	Parabolic cylinder	Parallel to y -axis.
$x^2 + y^2 = 4$	Circular cylinder	Parallel to z -axis

Contour diagrams

Consider $f(x, y) = 9 - x^2 - y^2$.



Take horizontal slice of the graph of f :

$$\{(x, y, z) : z = 9 - x^2 - y^2\} \cap \{(x, y, z) : z = a\}$$

Here, " a " is constant that we can move

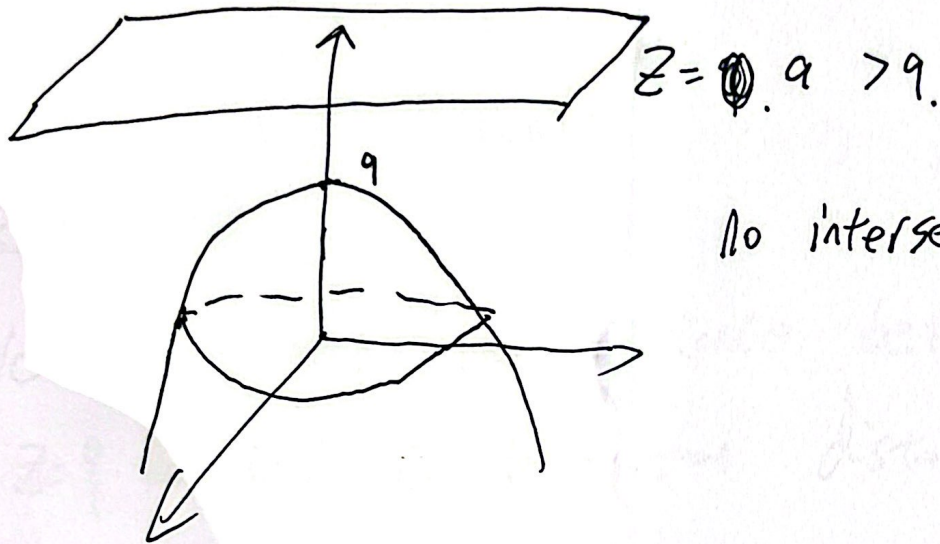
$$= \{(x, y, z) : z = 9 - x^2 - y^2, z = a\}$$

$$= \{(x, y, a) : x^2 + y^2 = 9 - a\}$$

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~~Since~~

Since $x^2 + y^2 \geq 0$, if $q - a < 0 \Leftrightarrow a > q$
then $x^2 + y^2 = q - a$ ~~does~~ Can't happen.

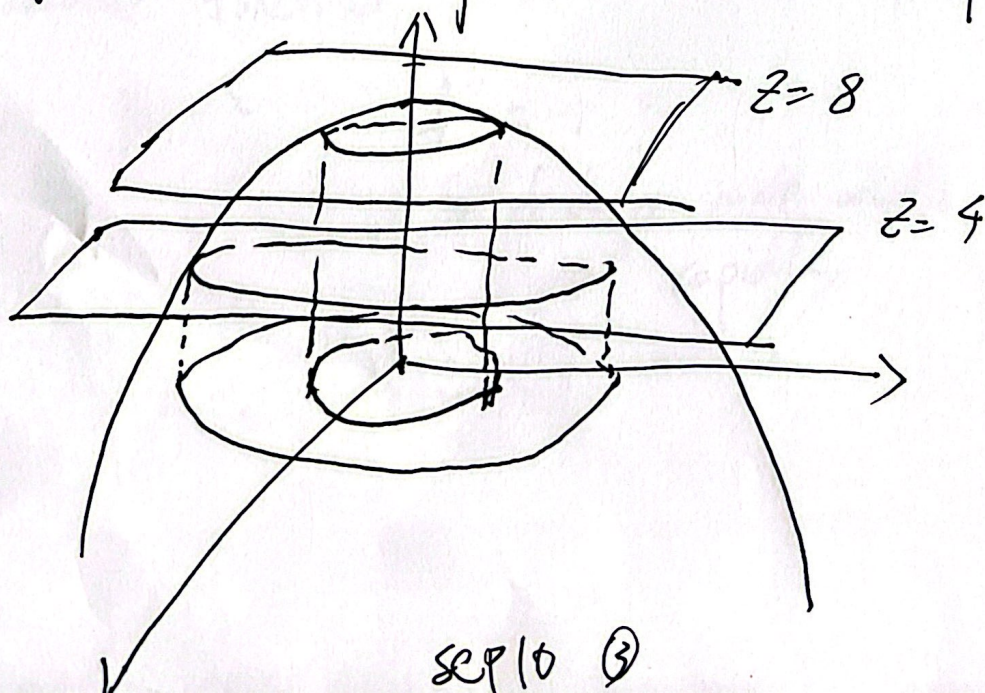


No intersection.

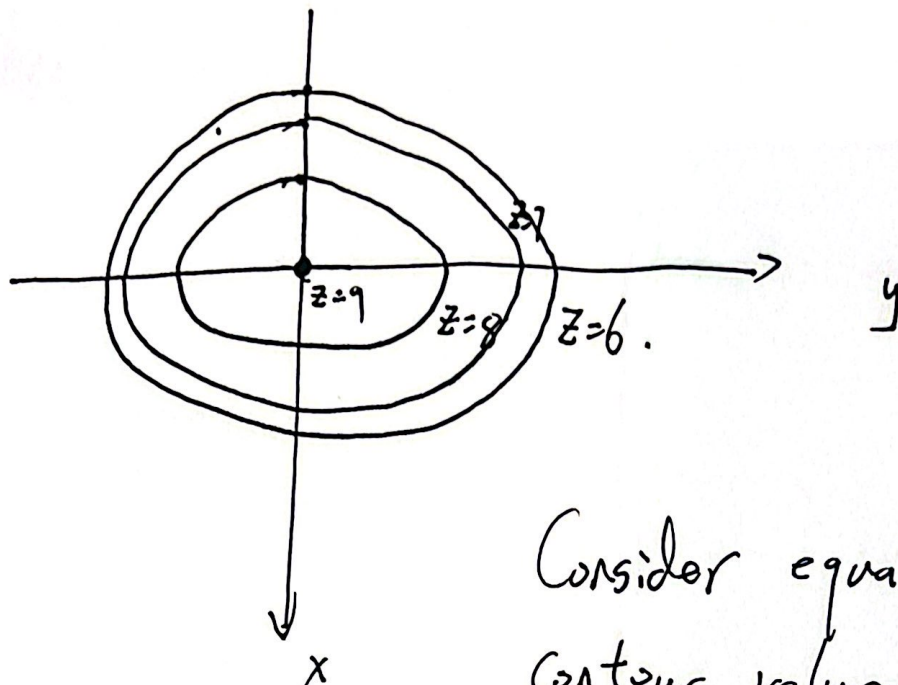
For $a \leq q$, $\{(x, y, a) : x^2 + y^2 = q - a\}$ is nonempty,

this is Contour diagram of $f(x, y) = q - x^2 - y^2$.

We can put Contour diagrams on a (x, y) -plane.



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~~This is called~~

Consider equally spaced.
Contour values: $z=9, 8, 7, 6, \dots$

We observe that the distance between $z=9$ and $z=8$ is bigger than distance between $z=8$ and $z=7$

This indicates that the function decay more and more rapidly.

Take one variable function as an example.

~~$f(x) = 9 - x^2$~~ $f(x) = 9 - x^2$, $f'' = -2 < 0$.

